

Page 6 – Useful Methods of Wrapper Classes

What is valueOf()?

Static method that converts a String/primitive into a Wrapper Object.

Example:

```
Integer obj=Integer.valueOf("100");
```

Returns: Integer object.

What is parseXxx()?

Static method that converts a String into a primitive.

```
int n=Integer.parseInt("100");
```

Returns: primitive int.

What is xxxValue()?

Instance method that converts Wrapper Object to primitive.

```
int x=obj.intValue();
```

Common Interview Questions

Difference between valueOf() and parseInt()? → valueOf returns object, parseInt returns primitive.

Why use valueOf()? Memory efficient and object required.

30-Second Answer

Wrapper classes provide valueOf(), parseXxx() and xxxValue() to convert between Strings, wrapper objects and primitives.

Page 7 – Access Modifiers

Definition

Access modifiers control visibility. Java provides private, default, protected and public.

Outer vs Inner Class

Outer class: only public/default.

Inner class: all four modifiers allowed.

Why?

Outer classes are top-level declarations. Inner classes are members like fields and methods.

Interview Questions

Can outer class be private? No.

Can inner class be private? Yes.

30-Second Answer

Top-level classes support only public/default while inner classes support all four modifiers.

Page 8 – Rules for Public Classes

Rule 1

Only one public class per Java file.

Rule 2

File name must match the public class name.

Rule 3

Only public classes are directly accessible from another package.

Interview Questions

Can a Java file have no public class? Yes.

Can it have multiple classes? Yes.

Page 9 – Instance Initialization Block

Definition

Runs every time an object is created and executes before the constructor.

Execution Order

Object Creation → Instance Initialization Block → Constructor

Important Points

Multiple blocks execute in order. If no constructor exists, Java provides a default constructor and the block still runs.

Interview Questions

Runs how many times? Once per object.

Can it use this? Yes.

Page 10 – Static Initialization Block

Definition

Runs only once when the JVM loads the class.

When It Executes

On first class loading: object creation, static method call or static field access.

Rules

Can directly access only static members. Cannot use this/super.

Execution Order

Static Block → Instance Block → Constructor

Interview Questions

How many times? Once.

Can static block access instance variables? No.

30-Second Answer

Static blocks initialize class-level resources once before any object is created.